ANALYZING SMART CITY BIKE SHARING DATA USING POWER BI

1. Project Overview

This project analyzes bike-sharing station data from multiple cities. The objective is to understand station availability, operational efficiency, bike utilization, and city-wise performance using Power BI. The project includes data cleaning, data modeling, DAX calculations, interactive dashboards, and business insights. The dataset contains bike station information such as station name, city, status, available bikes, bike stands, geographical position, and update timestamps.

Objective

The primary objective of this project is to analyze smart city bike-sharing data using Power BI and develop an interactive dashboard that provides insights into station performance, bike availability, utilization efficiency, and operational status across multiple cities. The project aims to support better decision-making through data cleaning, modeling, visualization, and business intelligence techniques.

2. Data Sources

- **Source Name:** Bike Sharing Station Dataset
- **Domain**: Transportation

3. Problem Statement

Public bike-sharing systems generate large amounts of operational data from stations located across different cities. Managing bike availability, station utilization, and operational performance can be challenging without proper analysis.

The objective of this project is to analyze bike-sharing station data using Power BI and provide actionable insights regarding bike availability, station performance, utilization efficiency, and city-wise operations.

4. Attribute (Column /Features) Details:

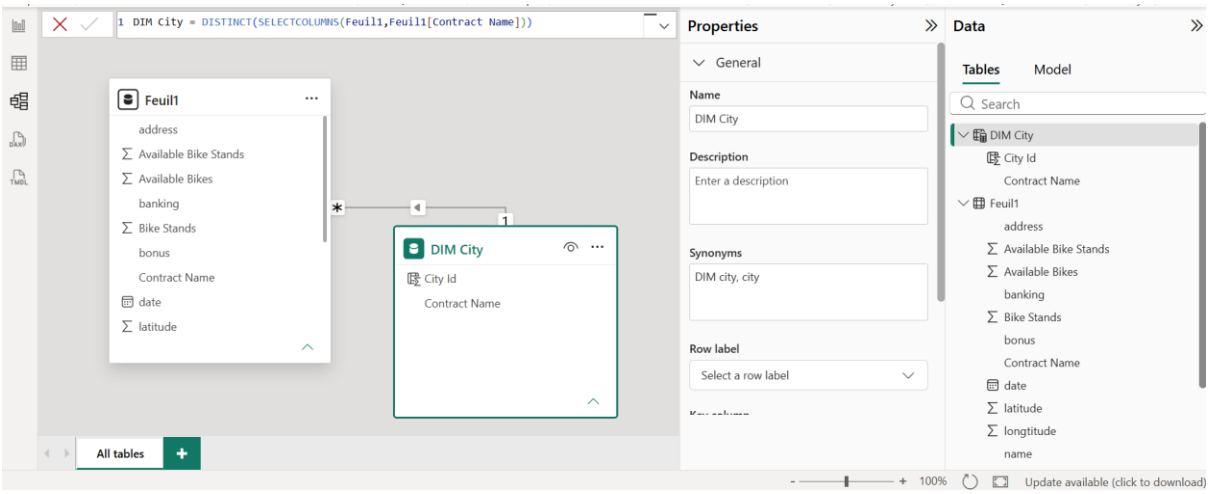
Attribute Name	Data Type	Description
Number	Whole Number	Unique identifier for each station
Name	Text	Name of the bike sharing station
Address	Text	Street Address
Position	Text	Geographical latitude and longtitude
Banking	Text	Indicating banking facility availability
Bonus	Text	Indicates whether station is a bonus station
Status	Text	Operational Status
Contract Name	Text	Operating contract name
Bike Stands	Whole Number	Total capacity of station
Available Bike Stands	Whole Number	Empty stands currently available
Available Bikes	Whole Number	Bikes currently available at station

5. Tools & Technologies

- Power BI Desktop: Importing data, creating measures, building visuals, and designing the report.
- Power Query: Checking column names and data types, cleaning data, and applying transformations.
- DAX (Data Analysis Expressions): Creating calculations for total stations, open/closed stations, bike stands, available bikes, and empty stands.

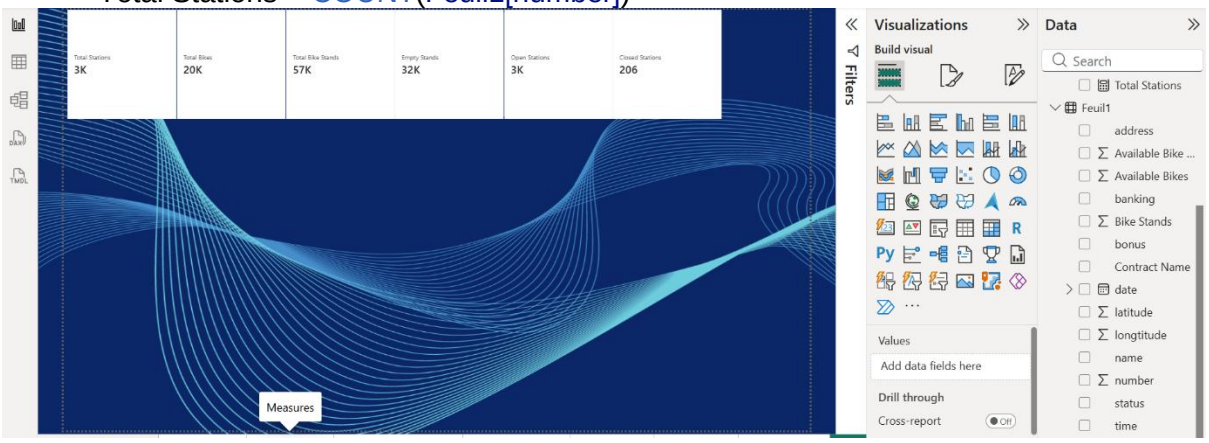
6. Data Modelling and DAX (Power BI)

Data Model:



DAX Measures:

- City Id = RANKX('DIM City','DIM City'[Contract Name], ,ASC,Skip)
- Closed Stations = CALCULATE(COUNTROWS(Feuil1),Feuil1[status]="CLOSED")
- DIM City = DISTINCT(SELECTCOLUMNS(Feuil1,Feuil1[Contract Name]))
- Empty Stands = SUM(Feuil1[Available Bike Stands])
- Open Stations = CALCULATE(COUNTROWS(Feuil1),Feuil1[status]="OPEN")
- Total Bike Stands = SUM(Feuil1[Bike Stands])
- Total Bikes = SUM(Feuil1[Available Bikes])
- Total Stations = COUNT(Feuil1[number])



7. Analysis and Visualizations (Power BI)

Dashboard Features:

• Card – Total Stations

• Card – Open Stations

• Card – Closed Stations

• Card – Total Bike Stands

• Card – Total Bikes

• Card – Empty Stands

• Donut chart – Open vs Closed Stations

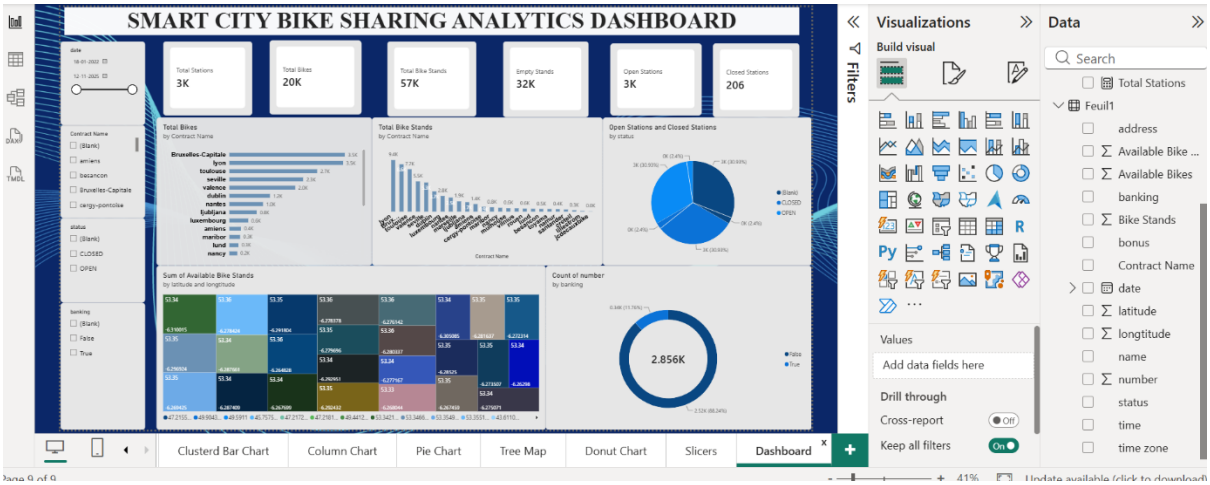
• Bar chart – Station count by Contract Name (city)

• Clustered column chart – Available Bikes and Available Bike Stands by city

• Table/Matrix – City-wise station status and availability details

• Map visual – Station locations using latitude and longitude (if coordinates are separated)

• Slicers – Contract Name, status, banking, and bonus



8. Insights:

Key Findings:

- Total station records: 2,856
- Open stations: 2,650
- Closed stations: 206
- City/contract names: 25
- Total bike stands: 57,203
- Available bikes: 19,646
- Available empty stands: 32,118
- City with the highest station count: Lyon (448 stations)
- Second-highest station count: Toulouse (431 stations)
- Marseille has 52 closed station records out of 130 station records

Descriptive Analysis:

The dataset contains 2,856 station records across 25 city/contract names. Of these, 2,650 are marked OPEN and 206 are marked CLOSED. The network records 57,203 bike stands, 19,646 available bikes, and 32,118 empty stands.

Diagnostic Analysis:

Lyon has the highest number of station records, followed by Toulouse. Marseille shows 52 closed station records out of 130

Predictive Analysis:

This dashboard describes the recorded station availability. It does not provide a formal forecast because historical repeated observations are not available. With regular time-based data, future demand and availability can be predicted more accurately.

Prescriptive Analysis:

- Monitor stations with very few available bikes and stations with very few empty stands; consider rebalancing after checking local demand.
- Use city and station filters to prioritize operational checks.
- Validate unusual capacity and availability values and refresh the source data regularly.
- Collect trip history, rentals, maintenance details, and repeated timestamps to support more reliable rebalancing decisions.

9. Conclusion:

This project demonstrates an end-to-end bike-sharing station analysis workflow using Excel and Power BI. The source data is prepared in Power Query, DAX measures are created to calculate station status and availability indicators, and an interactive dashboard is designed to explore city-wise performance. Descriptive and diagnostic analysis provide a clear operational overview, while predictive analysis is identified as a future enhancement because the dataset is snapshot-based. Prescriptive recommendations focus on station review, availability monitoring, and improved data collection. The dashboard supports evidence-based discussion and practical decision-making in smart-city transportation.